



DNA MODIFYING ENZYMES USED IN BIOTECHNOLOGY –LATEST UPDATES

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ABSTRACT

With the widespread adoption of molecular cloning as a routine laboratory technique, there is now a vast array of enzyme sources available from manufacturers for the generation and manipulation of nucleic acids. However, for those new to the field, the task of selecting the most suitable enzyme for a specific task can be daunting. This review aims to provide readers with valuable insights into the function and specificities of various sources of polymerases, ligases, nucleases, phosphatases, methylases, and topoisomerases commonly used in molecular cloning. We offer a comprehensive description of the most frequently utilized enzymes within each group, along with an explanation of their properties and mechanisms of action. By highlighting the key requirements for each enzymatic activity and addressing their limitations, our objective is to assist readers in making informed decisions when selecting the appropriate enzymatic source and determining the optimal experimental conditions for their molecular cloning experiments. Key words: molecular cloning, polymerases, ligases, nucleases, phosphatases, methylases, and topoisomerases

KEYWORDS: Molecular Cloning, Polymerases, Ligases, Nucleases, Phosphatases, Methylases, And Topoisomerases

INTRODUCTION

Molecular cloning involves the use of nucleic acids, which can be either natural or synthetic and vary in length from a few to several thousand nucleotides.

These nucleic acids can be manipulated extensively to acquire specific characteristics and properties, including propagation, ligation, digestion, and the addition of modifying groups such as phosphate or methyl groups.

These modifications are catalyzed by various enzymes, including polymerases, ligases, nucleases, phosphatases, and methylases. In this review, we aim to provide a comprehensive description of the main enzymes in each group, including their properties and mechanisms of action.

According to T. A. Brown here are 4 main enzyme (Brown, 2020).

- 1) **DNA polymerases**
- 2) **Nucleases**
- 3) **Ligases**
- 4) **End-modification enzymes**

- 1) **DNAPOLYMERASES**

The enzymes responsible for synthesizing new polynucleotides that are complementary to an existing DNA or RNA template are commonly referred to as DNA polymerases. These enzymes are capable of adding nucleotides solely to the 3'-OH end of a pre-existing primer that contains a 5'-phosphate group. In order to initiate DNA synthesis, short stretches of RNA that are complementary to approximately 10 nucleotides of DNA at the 5' end of the molecule are required. These RNA primers are synthesized by an RNA polymerase known as primase.

THE T4 BACTERIOPHAGE polymerase, also known as T4 DNA polymerase,

requires both a template and a primer to exhibit two distinct activities. In the absence of deoxynucleoside triphosphates (dNTPs), it functions as a 3'→5' (reverse) exonuclease. Conversely, in the presence of dNTPs, it acts as a 5'→3' polymerase. Unlike *E. coli* DNA Pol I, *T4 DNA Pol* does not possess a 5'→3' exonuclease activity (Rit-tié and Perbal, 2008). As a result, T4 DNA Pol can be utilized in lieu of Klenow for filling in 5'-protruding ends of DNA fragments, nick translation, or labeling 3' ends of double-stranded DNA. The T4 DNA Pol exonuclease rate is approximately 40 bases per minute on double-stranded DNA and about 4,000 bases on single-stranded DNA. Under standardized conditions, the T4 DNA Pol polymerization rate reaches 15,000 nucleotides per minute.

Richardson et al. (Tabor and Richardson, 1989) have reported on a chemically modified bacteriophage T7 DNA polymerase that is highly suitable for DNA sequencing. This modified T7 DNA polymerase has an impressive polymerization rate of over 300 nucleotides per second, making it more than 70 times faster than the AMV reverse transcriptase (**Avian Myeloblastosis Virus (AMV) Reverse Transcriptase** is an RNA-directed DNA polymerase.

DNA synthesis from either RNA or ssDNA is possible with AMV and it is ideal for RT-PCR, cDNA synthesis, and RNA sequencing. By conducting **in-vitro mutagenesis** on the corresponding nucleotides in the T7 gene 5, the exonuclease activity has been completely eliminated, resulting in a significant increase in polymerase efficiency (9-fold) and spontaneous mutation rate (14-fold). This mutant T7 polymerase/thioredoxin complex, commercially known as sequenase (**United States Biochemical Corporation**), is widely used for DNA sequencing due to its exceptional efficiency and ability to incorporate nucleotide analogs. These analogs, such as 5'-(α -thio)-dNTPs, dc7-

Table-1 Main DNA polymerases used in molecular biology (Rittié and Perbal, 2008).



Enzyme	Requirements	Activity	Main applications/notes
<i>Prokaryotic DNA polymerases</i>			
Pol I (removal of RNA primer from the 5' end of DNA chains)	Template	5'→3' polymerase	DNA replication
	Primer	3'→5' exonuclease (proofreading)	Nick translation (The process of tagging entails replacing specific nucleotides in a DNA sequence with labeled analogues.)
	dNTPs	5'→3' exonuclease	Removal of 3' protruding DNA ends (without dNTPs) Primer removal
Pol II	Template	5'→3' polymerase	DNA replication
	Primer	3'→5' exonuclease	
	dNTPs	exonuclease	
Pol III (Kornberg enzyme)	Template	5'→3' polymerase	DNA replication 3'→5' exonuclease (proofreading) activities of DNA Pol I while the small fragment exhibits the 5'→3' exonuclease activity alone.
	Primer	3'→5' exonuclease	
	dNTPs	exonuclease	
Klenow (large fragment of Pol I)	Template	5'→3' polymerase	DNA replication when exonuclease activity in 3' needs to be avoided (fill in large gaps)
	Primer	3'→5' exonuclease	
	dNTPs	exonuclease	
T4 DNA Pol	Template	5'→3' polymerase (with dNTPs)	DNA replication when exonuclease activity

Activate W

(A) DNA polymerases



Figure 1 Function of DNA polymerase

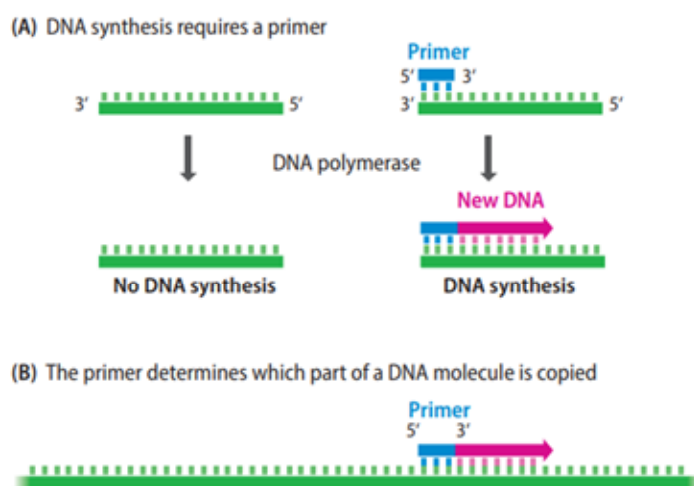


Figure 2 : Mechanism of DNA synthesis

GTP, or dITP, are employed to improve the resolution of DNA sequencing gels and prevent gel compression caused by base pairing. Taq polymerase was purified by Chien et al. in 1976 from *Thermophilus aquaticus*, a bacterium that was discovered in 1969 in the Great Fountain region of Yellowstone National Park by Brock and Freeze. *T. aquaticus* is able to thrive at temperatures of 70°C and can survive at even higher temperatures of up to 80°C. The half-life of Taq polymerase is 40 minutes at 95°C and 5 minutes at 100°C, making it suitable for use in the polymerase chain reaction (PCR). To achieve optimal activity, Taq polymerase requires the presence of Mg²⁺ and a temperature of 80°C. Although Taq polymerase does not possess 3'→5' exonuclease activity, which is responsible for proofreading and error correction, it is still considered a low fidelity enzyme with an error rate ranging from 1×10^{-4} to 2×10^{-5} errors per base pair, depending on the specific experimental conditions.

However, it is worth noting that this error rate still corresponds to a relatively high level of accuracy, as approximately 45,000 nucleotides can be incorporated into newly synthesized DNA strands before an error occurs. Similar to other DNA polymerases that lack 3'→5' exonuclease activity, Taq polymerase

exhibits a deoxynucleotidyl transferase activity, which involves the addition of a few adenine residues at the 3'-end of PCR products. Thermostable DNA polymerases with proofreading activity are enzymes isolated from thermophilic organisms that exhibit 3'→5' exonuclease activity, which increases fidelity. Among them, the DNA polymerase from the archaeobacterium *Pyrococcus furiosus* (Pfu) has an error rate (1.6×10^{-6}) 10-fold lower than that of Taq polymerase and can be purchased from numerous providers.

Pow polymerase, isolated from *Pyrococcus woesei*, has a half-life greater than 2 hours at 100°C and a very low error rate of 7.4×10^{-7} . Pow polymerase provides high PCR yields only for templates shorter than 3.5 kb.

To circumvent PCR product size limitation, a mixture of Taq and Pow DNA polymerases has been commercialized by Roche Molecular Chemicals (under the name “Expand high fidelity”). This system allows amplification and cloning of long stretches of DNA (20–35 kb) and represents a unique tool for analyzing and sequencing large eukaryotic genes or introducing large DNA fragments in lambda phages or cosmid vectors.

Vent polymerase, also known as Tli polymerase since isolated from *Thermococcus litoralis*, has an error rate comprised between that of Taq and Pfu. Vent polymerase has a half-life of 7 hours at 95°C and is marketed by New England Biolabs, together with a modified version that lacks exonuclease (proofreading) activity. Two DNA polymerases (Pol I and II) that exhibit 3'→5' exonuclease proofreading activity were also isolated from *Pyrococcus abyssi* (Gueguen et al., 2001; Zappa et al., 2001). Pol I DNA polymerase from *P. abyssi* (Pab), marketed by Qbiogen as “is DNA Polymerase”, has an error rate (0.66×10^{-6}) similar to that of Pfu.

However, Pab is more thermostable than Plu or Taq (Pab has a half-life of 5 hours at 100°C), making it very useful for conducting PCR reactions that involve high-temperature incubations.

2. REVERSE TRANSCRIPTASE

Which is an RNA-dependent DNA polymerase and so makes DNA copies of RNA rather than DNA templates. Reverse transcriptase are involved in the replication cycles of retroviruses. Including the human immunodeficiency viruses, which have RNA genomes that are copied into DNA after infection of the host. In the test tube, a reverse transcriptase can be used to make DNA copies of mRNA molecules. These copies are called complementary DNAs (cDNAs). Their synthesis is important in some types of gene cloning and in techniques used to map the regions of a genome that specify particular mRNAs. Because RTs are deprived of 3'→5' exonuclease (proofreading) activity, they have much lower fidelity than DNA-dependent DNA polymerases. For instance, HIV RT has an error rate of

1 mutation per 1,500 nucleotides, whereas RT of avian and murine origin generate 1 mutation per 17,000 and 30,000 nucleotides, respectively. The AMV/MAV RT is comprised of two subunits, α and β , which are structurally related. The α subunit weighs 65 kDa and the β subunit weighs 95 kDa. The enzyme's α subunit provides both RT and RNase H activities. The RT activity requires the presence of a primer and a template, as noted by Leis et al. (1983) and Verma (1977). The AMV/MAV RT is commonly used to replicate total messenger RNAs using polydT or random primers. (Verma, 1977). The RNase H activity is produced by the proteolytic cleavage of the α subunit and is associated with a 24 kDa fragment. RNase H is a processive exoribonuclease that specifically degrades RNA strands in RNA-DNA hybrids in either the 5'→3' or 3'→5' directions.

Uses

1) in Reverse Transcription Polymerase Chain Reaction (RT-PCR). 2)

3. NUCLEASES

Nucleases are a class of enzymes that possess the ability to cleave, shorten, or degrade nucleic

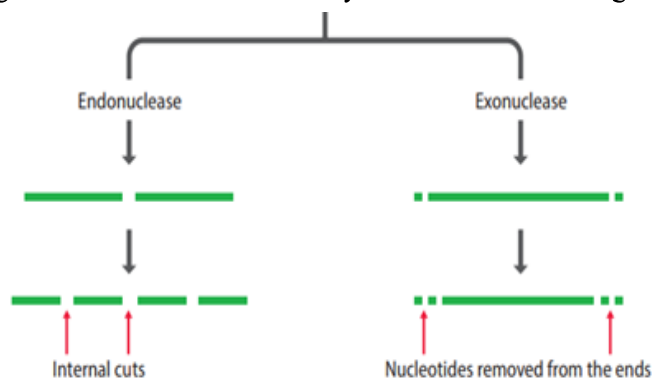
Table 2 Reverse transcriptases (Rittié and Perbal, 2008)

Reverse transcriptases			application
AMV/MAV RT	RNA Template	RT	Synthesis of cDNA from RNA (RT-PCR)
	Primer	RNase H	
MuLV RT	RNA Template	RT	RT-PCR for long transcripts
	Primer	RNase H	
Thermostable reverse transcriptases			
Pfu	Template	RT in the presence of Mn ²⁺	PCR or RT-PCR, depending on buffer composition
	Primer	DNA Pol in the presence of Mg ²⁺	
	Mn ²⁺		
MonsterScript™ RT	Template	RT, no RNase H activity	RT-PCR for long transcripts
	Primer		
Klenow fragment of <i>C. therm</i>	Template	RT	RT-PCR
	Primer		
	Mg ²⁺		

Reverse transcriptases			application
AMV/ MAV RT	RNA Tem- plate	RT	Synthesis of cDNA from RNA (RT-PCR)
	Primer	RNase H	
MuLV RT	RNA Template	RT	RT-PCR for long transcripts
	Primer	RNase H	
Thermostable reverse transcriptases			
Tth	Template	RT in the presence of Mn ²⁺	PCR or RT-PCR, depending on buffer composi- tion
	Primer Mn ²⁺	DNA Pol in the presence of Mg ²⁺	
Monster- Script™ RT	Template, Primer	RT, no RNase H activity	RT-PCR for long transcripts
Klenow fragment of <i>C. therm</i>	Template, Primer, Mg ²⁺	RT	RT-PCR

acid molecules. There are two distinct types of nuclease enzymes: exonucleases and endonucleases.

Which degrade DNA molecules by break-



ing the phosphodiester bonds that link one nucleotide to the next (Figure .3.) Which degrade DNA molecules by breaking the phosphodiester bonds that link one nucleotide to the next (Figure .3.)

Nucleases are enzymes that hydrolyze phosphodiester bonds in the backbone of nucleic acids.

These enzymes can be classified into two main categories based on their mode of action.

Exonucleases act on the ends of nucleic acid molecules, while endonucleases cleave nucleic acids internally.

Deoxyribonucleases specifically cleave DNA and produce nicks, which are points in a double-stranded DNA molecule where there is no phosphodiester bond between adjacent nucleotides of one strand.

These nicks can occur due to damage or the action of enzymes. On the other hand, ribonucleases cleave RNA. Nucleases play a dual role for molecular biologists.

On one hand, they pose a significant threat to the integrity of nucleic acids. On the other hand, they are invaluable tools for cutting and manipulating nucleic acids for cloning purposes.

S1 nuclease (EC 3.1.30.1) is a valuable tool for measuring the extent of hybridization between

DNA and RNA, probing duplex DNA regions, and removing single-stranded DNA of protruding ends generated by restriction enzymes. This enzyme is purified from *Aspergillus oryzae* and degrades RNA or single-stranded DNA into 5' mononucleotides, while not degrading duplex DNA or RNA-DNA

hybrids in their native conformation. S1 nuclease is a 32 kDa metalloprotein (Vogt 1973) that requires Zn^{2+} for activity. Although Co^{2+} and Hg^{2+} can replace Zn^{2+} , they are less effective as cofactors. The optimal activity of S1 nuclease is achieved at pH 4.0-4.3, with a 50% reduction of activity observed at pH 4.9, and the enzyme becoming inactive at pH > 6.0.

Chelating agents such as EDTA and citrate, or low concentrations of sodium phosphate (as low as 10 mM), strongly inhibit S1 nuclease. However, this nuclease is resistant to denaturing agents such as urea, SDS, or formamide. S1 nuclease exhibits a hydrolytic activity that is five times more rapid on single-stranded DNA compared to RNA, and an astonishing 75,000 times faster on double-stranded DNA. The minimal occurrence of strand breaks induced by S1 nuclease in duplex DNA can be further diminished by employing a high salt concentration, specifically 0.2 M. S1 nuclease demonstrates its enzymatic efficacy at S1 sensitive sites that arise from negative supercoiling of the helical DNA structure, UV irradiation, or depurination (Wiegand, Godson and Radding, 1975; Kroeker, Kowalski and Laskowski, 1976).

APPLICATION OF S1 NUCLEASE

- 1) For specific removal of single portions of duplex DNA, RNA/DNA hybrids, or RNA molecules.
- 2) mapping of spliced RNA molecules, isolation of duplex regions in single stranded viral genomes.
- 3) probing strand breaks in duplex DNA molecules.
- 4) mapping of the genomic regions involved in interactions with DNA binding proteins.

ASE is a highly specific single-stranded DNA and RNA endonuclease that has been purified from mung bean sprouts. This enzyme produces mono- and oligonucleotides with 5'-phosphate terminations. To achieve maximum activity and stability, mung bean nuclease requires the presence of Zn^{2+} and a reducing agent such as cysteine. However, it is important to note that high salt concentrations, specifically in the range of 200-400 mM NaCl, can inhibit its activity by 80-90%. To enhance the stability of mung bean nuclease and prevent its adhesion to surfaces, the addition of Triton X-100 at a low concentration of 0.001% w/v, particularly when used at less than 50 U/ μ l, can be beneficial. Mung bean nuclease can be employed similarly to nuclease S1 for the purpose of eliminating protruding ends in duplex DNA or for transcription promoter mapping (Kroeker, Kowalski and Laskowski, 1976).

4. LIGASES

Ligases, join DNA molecules together by synthesizing phosphodiester bonds between nucleotides at the ends of two different molecules or at the two ends of a single molecule. The enzyme used most often in experiments is T4 DNA ligase, which is purified from *E. coli* cells infected with bacteriophage T4.

DNA ligase is an important cellular enzyme, as its function is to repair broken phosphodiester bonds that may occur at random or as a consequence of DNA replication or recombination.



Enzyme	Requirements	Activity	Main applications/notes
DNA ligases			
T4-DNA Ligase	Mg ²⁺ , /ATP	Connects blunt and cohesive ends in duplex DNA, RNA or DNA/ RNA hybrids	Most frequently used for cloning
E. Coli DNA ligase	Mg ²⁺ / NAD ⁺	Connects preferentially cohesive double-stranded DNA ends, active on blunt ends DNA in the presence of Ficoll or polyethylene glycol	Ligation when blunt end or RNA/ DNA ligation needs to be avoided
Thermostable DNA ligases (various sources)		Ligation at high temperature	Not a substitute for T4 or E. Coli ligases, but used for specific techniques like LCR.
RNA ligases			
T4 RNA ligase 1	ATP	Ligates single stranded nucleic acids and polynucleotides to RNA molecules	RNA labeling, primer extension
T4 RNA ligase 2 (T4 Rnl-2)	ATP	Ligates double-stranded RNA or connects dsRNA to dsDNA	Repair nicks in dsRNA
Truncated T4 RNL2		Ligates pre-adenylated 5' end of DNA or RNA to 3' end of RNA molecules	Optimized linker ligation for cloning of microRNAs

In cellular biology, DNA ligases play a crucial role in the joining of Okazaki fragments during replication, as well as in the final stage of the DNA repair process. In molecular biology, DNA ligases are utilized to connect DNA fragments with either blunt or sticky ends, which are generated by restriction enzyme digestion, to add linkers or adaptors to DNA, or to repair nicks.

The operation of DNA ligases involves a three-step reaction. The first step involves the creation of a ligase-adenylate intermediate, in which a phosphoamide bond is formed between a lysine residue and one AMP molecule of the enzyme cofactor (ATP or NAD⁺). The second step involves the transfer of the AMP to the 5'-phosphate end of the DNA nick to form a

DNA-adenylate (AppDNA). Finally, a nucleophilic attack from the 3' end of the DNA nick directed to the AppDNA results in the joining of the two polynucleotides and the release of AMP. Bacterial DNA ligases utilize NAD⁺ as a cofactor, while DNA ligases from eukaryotes, viruses, and bacteriophages use ATP. The most commonly used ligase in molecular biology is the bacteriophage T4 DNA ligase, which is a 68 kDa monomer that requires Mg²⁺ and ATP as cofactors. T4 DNA ligase is capable of connecting blunt and cohesive ends, as well as repairing single-stranded nicks in duplex DNA, RNA, or DNA/RNA hybrids.

The *E. Coli* DNA ligase exhibits a preference for cohesive double-stranded DNA ends. How-

ever, it can also function on blunt ends DNA in the presence of Ficoll or polyethylene glycol. Hybrids such as DNA-RNA or RNA-RNA are not efficiently formed by *E. Coli* DNA ligase. This characteristic can be advantageous when double-stranded DNA ligation is desired and blunt end ligation needs to be avoided.

Thermostable DNA ligases possess the capability to conduct ligation of duplex molecules and repair single stranded nicks within a temperature range of 45 to 80°C. These ligases exhibit a high level of specificity and are particularly suitable for applications requiring stringent ligations. They are derived from various sources, including *Thermus thermophilus*.

Thermostable DNA ligases are typically not employed as replacements for T4 or *E. Coli* DNA ligases, but rather find utility in highly specific methodologies such as Ligase Chain Reaction (LCR).

LCR serves as a technique for identifying single base mutations, wherein a primer is synthesized in two segments that encompass both ends of a potential mutation. The thermostable ligase will solely join the two segments if they precisely match the template sequence. Subsequent PCR reactions will only amplify if the primer is ligated.

RNA ligases (EC 6.5.1.3) are enzymes that facilitate the ATP-dependent synthesis of phosphodiester bonds between the 5'-phosphate and 3'-OH termini of single stranded RNA or DNA molecules. Their primary function in cellular processes is to repair damaged RNA molecules. Similar to DNA ligases, RNA ligases undergo a three-step reaction process (Silber et al. 1972; Sugino et al. 1977). Initially, the RNA ligase reacts with ATP to form a covalent intermediate known as ligase-(lysyl-N)-AMP, along with the release of pyrophosphate. Subsequently, the AMP component of the ligase adenylate is transferred to the 5'-phosphate end of the RNA, resulting in the formation of an RNA-adenylate intermediate (AppRNA). Finally, the 3'-OH end of the RNA undergoes a nucleophilic attack on

the AppRNA, leading to the creation of a phosphodiester bond that effectively seals the two RNA ends together.

5) END-MODIFICATION ENZYMES

End-modification enzymes, which make changes to the ends of DNA molecules

There are numerous enzymes that modify DNA molecules by addition or removal of specific chemical groups.

A). ALKALINE PHOSPHATASE

- This enzyme removes phosphate group at the 5' terminus of a DNA molecule.

The enzymes alkaline phosphatase, polynucleotide kinase, and terminal transferase play crucial roles in DNA manipulation by acting on the termini of DNA molecules. The alkaline phosphatase and polynucleotide kinase enzymes are responsible for the removal or addition of phosphate groups.

Bacterial alkaline phosphatase, along with its similar counterpart calf intestinal alkaline phosphatase, specifically removes phosphate groups from the 5 ends of DNA, resulting in a 5-OH group. This enzymatic activity is particularly useful in preventing unwanted ligation of DNA molecules, which can be problematic in certain cloning procedures. Additionally, it is employed prior to the addition of radioactive phosphate to the 5 ends of DNAs by polynucleotide kinase.

B) TERMINAL TRANSFERASE

Terminal deoxynucleotidyl transferase (TdT) is a DNA polymerase that was first isolated from calf thymus by Krakow et al. in 1962 (Krakow, Coutsogeorgopoulos and Canellakis, 1962). This enzyme catalyzes the addition of a homopolymer tail to the 3'-OH ends of DNA in a template-independent manner. TdT is commonly used in molecular biology for labeling DNA 3'

ends with modified nucleotides, such as ddNTP, DIG-dUTP, or radiolabeled nucleotides, as well as for primer extension and DNA sequencing. Additionally, TdT is utilized in the TUNEL (TdT dUTP Nick End Labeling) assay to demonstrate apoptosis, as described by Gavrieli et al. in 1992 (*Gavrieli, Sherman and Ben-Sasson, 1992; Ashley, Potts and Olorunniji, 2023*). Terminal deoxynucleotidyl transferase (TdT) is an enzyme that facilitates the incorporation of arbitrary nucleotides onto single-stranded DNA without the need for a template. This unique characteristic has made TdT a valuable tool in both biotechnology and clinical applications. One particularly intriguing application is the synthesis of long de novo DNA molecules through TdT-mediated iterative incorporation of a reversibly blocked nucleotide at the 3' end, followed by deblocking.

However, the wild-type (WT) TdT is not well-suited for the incorporation of modified nucleotides at the 3' end, and the engineering of TdT is hindered by its marginal stability and limited presence in mesophilic organisms. (Chua et al., 2020).

c) Polynucleotide kinase is an enzyme that catalyzes the addition of a phosphate group to the 5' terminus of a DNA molecule.

The T4 polynucleotide kinase (Pnk) is not only a valuable research tool, but also represents a group of bifunctional enzymes that possess both 5'-kinase and 3'-phosphatase activities, which are crucial in the repair of RNA and DNA. (Wang, Lima and Shuman, 2002).

6. Exonucleases are enzymes capable of breaking phosphodiester bonds at the terminal end of a DNA molecule.

9. An example of an exonuclease is Bal31, which is known for its ability to remove nucleotides from both sides of a double-stranded DNA molecule. Bal31 is purified from the bacterium *Alteromonas espenjiana*.

10. Exonuclease III

- It degrades just one strand of double stranded

DNA molecule and leave single stranded DNA as the product.

- 1) Bal 31
- 2) exonuclease III (exonucleases),
- 3) deoxyribonuclease I (DNase I)

Nuclease Bal 31 is a complex enzyme.

Its primary activity is a fast-acting 3' exonuclease, which is coupled with a slow-acting endonuclease. When Bal 31 is present at a high concentration these activities effectively shorten DNA molecules from both termini.

- 1) Exonuclease III is a 3' exonuclease that generates molecules with protruding 5' termini.
- 2) DNase I cuts either single-stranded or double-stranded DNA at essentially random sites.
- 3) Nuclease S1 is specific for single-stranded RNA or DNA.

11. 2). Endonuclease

- It is able to break internal phosphodiester bond of the DNA molecule.

12. S1 endonuclease

- It cleaves only single strand.
- It is purified from the fungus *Aspergillus oryzae*

13. Deoxyribonuclease 1 (Dnase 1)

- It cleaves both single stranded and double stranded DNA.
- It is purified from cow pancreas. This enzyme is non-specific.

14. RNase A :

It digests ssRNA at 3' terminus. • RNase H : It digests RNA on RNA-DNA heteroduplex.

15. SPECIAL CLASS OF ENDONUCLEASE

a) Restriction Endonuclease

This type of enzymes cleaves specific recognition site on double stranded DNA.

RESTRICTION ENDONUCLEASES

- 1) Also called restriction enzymes
- 2) 1962: "molecular scissors" discovered in bacteria
- 3) E. coli bacteria have an enzymatic im-

immune system that recognizes and destroys foreign DNA

4) 3,000 enzymes have been identified, around 200 have unique properties, many are purified and available commercially.

there are two different kinds of restriction enzymes:

(1) Exonucleases catalyses hydrolysis of terminal nucleotides from the end of DNA or RNA molecule either 5' to 3' direction or 3' to 5' direction. Example: exonuclease I, exonuclease II etc.

(2) Endonucleases can recognize specific base sequence (restriction site) within DNA or RNA molecule and cleave internal phosphodiester bonds within a DNA molecule.

Example: EcoRI, Hind III, BamHI etc.

Restriction Endonuclease Nomenclature:

Restriction endonucleases are named according to the organism in which they were discovered, using a system of letters and numbers. For example, HindIII (pronounced "hindee-three") was discovered in *Haemophilus influenza* (strain d). The Roman numerals are used to identify specific enzymes from bacteria that contain multiple restriction enzymes indicating the order in which restriction enzymes were discovered in a particular strain.

Named for bacterial genus, species, strain, and type

Example: EcoRI

Genus: *Escherichia*

Species: *coli*

Strain: R

Order discovered: 1

Enzymes recognize specific 4-8 bp sequences which are palindromic.

Uses of restriction enzymes

- RFLP analysis (Restriction Fragment Length Polymorphism)
- DNA sequencing
- DNA storage – libraries

- Transformation

- Large scale analysis – gene chips 2-1.4

CLASSIFICATION OF RESTRICTION ENDONUCLEASES:

There are three major classes of restriction endonucleases based on the types of sequences recognized, the nature of the cut made in the DNA, and the enzyme structure:

- Type I restriction enzymes
- Type II restriction enzymes
- Type III restriction enzymes

2-1.4.1 TYPE I RESTRICTION ENZYMES:

- These enzymes have both restriction and modification activities. Restriction depends upon the methylation status of the target DNA.

- Cleavage occurs approximately 1000 bp away from the recognition site.

- The recognition site is asymmetrical and is composed of two specific portions in which one portion contain 3–4 nucleotides while another portion contain 4–5 nucleotides and both the parts are separated by a non-specific spacer of about 6–8 nucleotides.

- They require S-adenosylmethionine (SAM), ATP, and magnesium ions (Mg^{2+}) for activity.

- These enzymes are composed of mainly three subunits, a specificity subunit that determines the DNA recognition site, a restriction subunit, and a modification subunit.

TYPE II RESTRICTION ENZYMES:

- Restriction and modification are mediated by separate enzymes so it is possible to cleave DNA in the absence of modification. Although the two enzymes recognize the same target sequence, they can be purified separately from each other.

- Cleavage of nucleotide sequence occurs at the restriction site.

- These enzymes are used to recognize rotationally symmetrical sequence which is often

referred as palindromic sequence.

- These palindromic binding site may either be interrupted (e.g. BstEII recognizes the sequence 5'-GGTNACC-3', where N can be any nucleotide) or continuous (e.g. KpnI recognizes the sequence 5'-GGTACC-3').
- They require only Mg²⁺ as a cofactor and ATP is not needed for their activity.
- Type II endonucleases are widely used for mapping and reconstructing DNA in vitro because they recognize specific sites and cleave just at these sites.

Cleaving a single piece of DNA with multiple restriction enzymes creates a "DNA fingerprint."

The pattern of fragments can be compared to similar DNA from another source treated with the same enzymes, to determine if the two are identical or different.

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